

Wireless Nurse Call System -WAid

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I. THE APPLICATION – WAID

The increasing staff shortage in healthcare hinders caretakers to adequately respond to patient requests and emergencies. To relieve overworked staff and ensure a fast and high quality patient care, scalable and energy efficient communication systems are necessary.

Classic wired systems are rigid, high maintenance, and only limited scalable. In contrast, with reliable and energy efficient routing algorithms, Wireless Sensor Networks (WSNs) can cover every room in healthcare facilities without a problem.

In this project the wireless nurse call system *WAid* based on the Routing Protocol for Low-Power and Lossy Networks (RPL) is introduced. The system aims to reliably record patient requests and emergencies by sorting them by urgency and notifying the nursing staff. Additionally, the patient's heart rate is continuously monitored to recognize critical events immediately.

II. BOARDS AND SENSORS

A. Application

From an application perspective, the system is divided into four board classes: Root, Access Points, Bed Nodes and Mobile Nurses.

Root. The Root initializes the network, maintains routing state for downward routing (Sec. III), receives emergency and assistance requests from the nodes (Sec. VI), and keeps a global overview of all active cases. Based on this overview, it prioritizes requests and forwards the next relevant task to the Mobile Nurse device (Sec. VII).

Access Points. Access Points act as forwarding nodes that extend the communication range of the Root and improve overall network coverage. Due to routing decisions Access Points can have limited processing capabilities.

Bed Nodes. Bed Nodes represent the patient-facing devices: they allow patients to submit requests via three dedicated buttons (emergency, toilet, and water). A fourth button is used to reset an active case to indicate that the nurse has taken care of it. In addition to call requests, each Bed Node periodically transmits the patient's heart rate obtained from an attached pulse sensor.

Mobile Nurse. Finally, the Mobile Nurse device serves as the staff interface and is used solely to notify the nurse about the next task or patient to attend to; it provides a buzzer-based alert that signals whenever a new prioritized task is assigned. The Mobile Nurse Device is expected to be highly mobile, and thus must adapt quickly to changing conditions.

B. Sensors

The Waid hardware integrates pulse sensors at each Bed Node to capture the patients heart rate. The measured signals are converted to BPM values and transmitted periodically to the network/gateway, enabling continuous monitoring and providing a basis for detecting abnormal conditions. This allows the system to react to changes and generate corresponding status updates.

For feedback at the mobile nurse device, the platform includes a buzzer that signals important events or alarms. Every time the emergency with the highest priority changes, the nurse gets notified.

The integrated buttons of our development boards are used to set and reset the alarm.

III. RPL – ROUTING PROTOCOL FOR LOW-POWER AND LOSSY NETWORKS

A. Protocol Overview

RPL is an Internet Protocol version 6 (IPv6) routing protocol specifically designed for Low-Power and Lossy Networks (LLNs), which are characterized by high loss rates, low data rates, and instability. Due to the device sizes and the unreliability of wireless links, WSNs are considered LLNs, making RPL a suitable routing protocol. RPL organizes the network into one or more RPL Instances, each forming a Destination-Oriented Directed Acyclic Graph (DODAG). Each DODAG originates in a DODAG Root, typically a gateway or border router. Within the DODAG, the nodes construct a tree structure by exchanging control messages, ensuring loop-free routing. The relative position of a node in the topology is expressed by its Rank, a scalar value that monotonically increases with the hops needed to reach the root [1]. The computation of the rank and the parent selection process are governed by an Objective Function explained in Sec. IV.

RPL supports two operating modes for downward routing: Storing Mode, where each node maintains routing tables for their subtrees, and Non-Storing Mode, in which the root maintains the complete routing information, making use of source routing [1], [2]. In WAid the Non-Storing mode is deployed. Since the Mobile Nurse is highly mobile Non-Storing mode ensures the root has the most recent routing information. Additionally, it allows for simpler intermediate node devices with less routing-state maintenance.

To reduce control overhead while maintaining responsiveness to topology changes, RPL employs an adaptive transmission strategy based on the Trickle principle described in

Sec. V and provides local and global repair mechanisms to restore a consistent and loop-free DODAG after link or node failures [1].

B. Control Messages

As mentioned above, control messages are exchanged between the nodes to build the DODAG. There are three control messages: DODAG Information Object (DIO), DODAG Information Solicitation (DIS), and Destination Advertisement Object (DAO). They are used to structure the DODAG, for upward and downward routing, and allow for a dynamic, self-healing topology [1].

DIO. The Root initializes a the network by constructing a DIO message and broadcasting it to the network. The DIO message includes the link layer address of the transmitter, the DIO version number, and the transmitter rank in the DODAG. The transmission of a DIO message is triggered by the trickle timer (Sec. V) with the transmission frequency depending on the stability of the network. Access Points, Bed Nodes, and the Mobile Nurse device receive these DIO messages and update their routing state by selecting a preferred parent. This decision is based on an objective function described in Sec. IV, and allows nodes to adapt to changing link conditions. To extend network coverage, Access Points additionally act as relays and forward DIO messages further into the network, enabling multi-hop connectivity for nodes that are outside the direct radio range of the Root. To ensure tight synchronization between the Root and Access Points, Access Points do not initialize their DIOs and version numbers [1].

DIS. Access Points, Bed Nodes, and the Mobile Nurse Device send DIS messages to rejoin the topology in case of loss of all parents or reset. To monitor the liveliness of parents a periodic timer is initiated in which reception of a DIO message from the parents is expected. If no DIO message is received during this period, the node solicits new DIO messages by broadcasting a DIS message. Therefore, DIS messages reduce downtime with unreachable nodes by accelerating reattachment to the network [1].

DAO. Finally, DAO messages are transmitted periodically by Access Points, Bed Nodes, and Mobile Nurse devices upwards via their parents to the Root. In the message the transmitter address is included, as well as the preferred parent address and a sequence number. Primarily, DAO messages advertise the node's current parent relation to the Root, allowing the Root to build a global routing table. Secondly, the Root checks the liveliness of nodes through DAO messages, removing dead nodes from the routing table if no DAO message was received in a certain period. DAO messages are forwarded by intermediate nodes without appending their own addresses to the payload. Instead, the Root constructs end-to-end paths from the collected parent-child relations. Consequently, the system follows the principle of Non-Storing Mode, where intermediate nodes are kept lightweight and routing state for downward paths is maintained at the Root [1].

IV. OBJECTIVE FUNCTION

As RPL is a distance-vector routing protocol, each node relies on an Objective Function to select a preferred parent. In WAid, parent selection is implemented using a composite scoring approach that combines the advertised rank of a candidate parent with a smoothed estimate of link quality derived from the Received Signal Strength Indicator (RSSI). Whenever an Access Point, Bed Node, or the Mobile Nurse device receives a DIO message, it updates its local candidate parent list using the received rank information and the corresponding smoothed RSSI value of the sender. The Objective Function then maps both metrics to normalized scores and combines them into a single composite value, which is used to select and maintain a preferred parent that provides both reliable connectivity and progress towards the root.

A. Exponential Moving Average of RSSI

To mitigate short-term fluctuations in RSSI measurements, WAid applies an exponential moving average filter to the RSSI values observed for each candidate parent i : The smoothing factor α is role-dependent and implemented as a power-of-two step size:

$$\alpha = 2^{-s}, \quad s \in \{1, 2\}. \quad (1)$$

B. Normalization of Rank and RSSI

For scoring, the advertised rank R_i of a candidate parent is mapped to a normalized rank score $S_R \in [0, 100]$ such that smaller ranks yield larger scores:

$$S_R(R_i) = \text{clip}_{[0,100]} \left(\frac{(R_{\max} - R_i) \cdot 100}{R_{\max}} \right), \quad (2)$$

where $R_{\max}$ denotes the configured maximum rank (RANK_MAX) and $\text{clip}_{[0,100]}(\cdot)$ saturates values to the interval $[0, 100]$.

The smoothed RSSI value is first clamped to a predefined range and then mapped linearly to a monotonic link-quality score $S_Q \in [0, 100]$:

$$\text{RSSI}_{c,i} = \text{clip}_{[\text{RSSI}_{\min}, \text{RSSI}_{\max}]}(\overline{\text{RSSI}}_i), \quad (3)$$

$$S_Q(\overline{\text{RSSI}}_i) = \text{clip}_{[0,100]} \left(\frac{(\text{RSSI}_{c,i} - \text{RSSI}_{\min}) \cdot 100}{\text{RSSI}_{\max} - \text{RSSI}_{\min}} \right), \quad (4)$$

where $\text{RSSI}_{\min}$ and $\text{RSSI}_{\max}$ correspond to RSSI_MIN_DBM and RSSI_MAX_DBM .

C. Composite Score and Parent Selection with Hysteresis

To obtain a stable topology that is robust against transient link variations, WAid computes a composite score for each candidate parent i as a weighted combination of the normalized rank score and the normalized link-quality score:

$$S_i = \frac{W_R \cdot S_R(R_i) + W_Q \cdot S_Q(\overline{\text{RSSI}}_i)}{100}, \quad (5)$$

where W_R and W_Q denote the configured weights (RANK_WEIGHT and RSSI_WEIGHT). A node switches from its current preferred parent p to a candidate parent i only if the

candidate offers a sufficiently large improvement, implemented via a hysteresis margin:

$$S_i > S_p + H, \quad (6)$$

where H is a hysteresis threshold that prevents frequent parent changes and thereby improves routing stability.

D. Multiple Parents

In RPL nodes keep a list of available parents and their connectivity. Based on the score calculated according to Sec. IV-C a preferred parent is chosen. If this parent becomes unavailable or the link quality worsens, the node switches to the second parent in the list. This mechanism allows fast parent switching without the need for sending a DIS, and thus minimizes periods of disconnection. Therefore, the network allows dynamic behavior without compromising robustness.

V. TRICKLE-BASED DIO DISSEMINATION

To efficiently distribute routing information along the DODAG while minimizing control overhead, WAid employs a Trickle-based timer mechanism for scheduling DIO messages. The Trickle algorithm dynamically adapts the transmission rate of these messages based on network stability. During steady-state operation, transmissions are sparse to reduce the load on the network during stable time periods. Topology changes result in more frequent transmissions, allowing a faster DODAG build. This behavior is well suited for LLNs, where energy efficiency and scalability are critical.

A. Trickle Interval Definition and Evolution

Let I_k denote the Trickle interval duration at interval index k . The initial interval is defined as

$$I_0 = I_{\min}. \quad (7)$$

The maximum interval is bounded by

$$I_{\max} = I_{\min} \cdot 2^d, \quad (8)$$

where d denotes the configured number of doublings.

At the end of each interval, the interval duration is doubled unless the maximum interval has been reached:

$$I_{k+1} = \min(2I_k, I_{\max}). \quad (9)$$

This exponential backoff reduces control traffic during stable network conditions.

B. Transmission Time Selection

Within each Trickle interval of length I_k , a single transmission time t_k is selected uniformly at random from the second half of the interval:

$$t_k \sim \mathcal{U}\left(\frac{I_k}{2}, I_k\right). \quad (10)$$

This randomization minimizes the probability of synchronized transmissions and collisions among neighboring nodes.

C. Trickle Reset and Topology Changes

Whenever the Root detects topology changes or receives a DIS, the trickle timer interval is reset. A reset forces the interval back to its minimum value. In addition, the DODAG version number is incremented to signal a new routing generation. This ensures that updated routing information is distributed quickly after topology changes.

VI. UPWARD MESSAGING (MSG_UP): RELIABLE EVENT AND HEARTBEAT DELIVERY

In WAid, application-layer events such as patient requests and periodic status updates (e.g., heart rate measurements) are delivered from Bed Nodes to the Root using an upward messaging mechanism denoted as MSG_UP. Messages are delivered along the DODAG graph, where each node passes the message to the selected parent, until it reaches the Root.

To increase reliability in case of unstable links and temporary parent loss, MSG_UP employs an acknowledgment-based reliability mechanism. After transmitting an upward message, the sender enters an acknowledgment-waiting state. If no acknowledgment is received within this predefined interval T_{ACK} , the message is retransmitted. The sender performs at most $N_{\max}$ retransmission attempts. If all retries are exhausted without successful acknowledgment, the transmission attempt is aborted.

A. Local Buffering under Parent Unavailability

When upward forwarding is temporarily not possible (e.g., due to the absence of a reachable preferred parent), messages are buffered locally in a bounded pending queue. Let $Q(t)$ denote the number of queued messages at time t , with a maximum capacity $Q_{\max}$. Queue occupancy evolves according to

$$Q(t^+) = \min(Q(t) + 1, Q_{\max}), \quad (11)$$

where messages are dropped if the queue is full. Once a parent becomes available again, buffered messages are forwarded in first-in-first-out order.

B. Emergency Processing and Prioritization at the Root

At the Root, received MSG_UP messages are processed according to their type and arrival time. Emergency messages are stored in an internal emergency table that contains all active cases and their prioritization. Heartbeat messages are displayed and if a pulse is beyond or below a certain threshold, an emergency is created. Let $\mathcal{E}$ denote the set of active emergencies. The Root maintains an ordered list

$$\mathcal{E} = \{(l_1, \pi_1), (l_2, \pi_2), \dots\}, \quad (12)$$

where l_i denotes the node label and π_i its associated priority, sorted such that $\pi_1 \leq \pi_2 \leq \dots$. This ordering enables the Root to determine which request should be forwarded next to the Mobile Nurse device.

C. Path-Based Acknowledgment Forwarding

After handling an upward message, the Root sends an acknowledgment back to the originating node. Because intermediate nodes do not maintain full downward routing state, acknowledgments are forwarded using an explicit path encoded by the Root. Let $(h_1, h_2, \dots, h_m)$ denote the hop sequence from the Root to the destination node. Each intermediate node forwards the acknowledgment to the next hop according to

$$\text{next_hop}_{\text{ACK}}(h_j) = h_{j+1}, \quad j = 1, \dots, m - 1. \quad (13)$$

When the acknowledgment reaches the destination node, the sender terminates its acknowledgment-waiting state and suppresses further retransmissions.

VII. DOWNWARD MESSAGING (MSG_DOWN): RELIABLE CONTROL DELIVERY TO THE NURSE DEVICE

Downward control communication in WAid is realized through the application-layer mechanism MSG_DOWN. Due to the operating Non-Storing mode, downward delivery requires explicit path guidance, as intermediate nodes do not maintain full downward routing state. MSG_DOWN reuses the same reliability concepts introduced for upward messaging, including end-to-end acknowledgments and retransmissions with bounded retry attempts.

The Root initiates MSG_DOWN transmissions mainly to the Mobile Nurse station to dispatch a new emergency task.

A. Path-Based Downward Forwarding

For a destination node d , the Root constructs an explicit hop sequence

$$P(d) = (h_1, h_2, \dots, h_m), \quad (14)$$

where h_1 is the first relay below the Root and $h_m = d$ is the final recipient. This path is part of the MSG_DOWN payload. Upon reception, each intermediate node forwards the message to the next hop specified in the sequence:

$$\text{next_hop}_{\text{DOWN}}(h_j) = h_{j+1}, \quad j = 1, \dots, m - 1. \quad (15)$$

Once the last hop is reached, the destination node applies the received control command locally (e.g., visual or acoustic notification of a new task).

B. Reliability Mechanism

Reliable delivery of MSG_DOWN follows the same acknowledgment and retransmission strategy as described for MSG_UP in Sec. VI. However, MSG_DOWN_ACK is forwarded by passing the message upwards to the preferred parent until it reaches the Root. Once the Root receives the acknowledgment, it marks the control delivery as successful and terminates any remaining retransmission attempts.

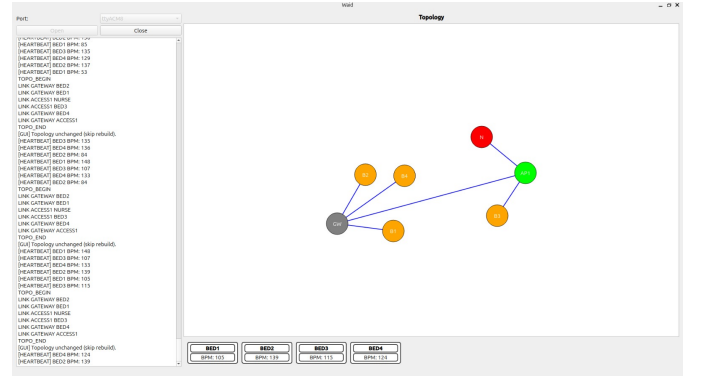


Fig. 1. Waid GUI showing serial port control and log output (left), as well as topology visualization and bed status indicators (right).

VIII. GRAPHICAL USER INTERFACE

The Waid application displays the routing topology and patient-related information. By presenting the network structure and real-time events, the interface contributes to reliable system operation.

The main window consists of two sections. The left section allows users to manage the serial port connection and view real-time log output. A drop-down menu lists available USB serial interfaces for selection. Incoming data is displayed in a read-only log that maintains a chronological record of events. This log enables users to verify system functionality, monitor alerts, and confirm heartbeat signals.

The right section presents a topology view with compact bed status displays. A QGraphicsView/QGraphicsScene pipeline is used to render devices as NodeItem objects and links as EdgeItem objects. Nodes are color-coded and labeled for identification. The gateway is shown in gray ("GW"), the access points in green ("APn"), bed nodes in orange ("Bn"), and nurse devices in red ("N"). Connections are depicted as lines behind nodes to reveal the network structure.

The topology layout updates are performed using a force-directed algorithm. The gateway remains fixed, and access points are aligned based on their names. Other nodes adjust their positions according to the preferred parent, with an underlying spring-like force mechanism to maintain appropriate spacing. If no topology changes occur, nodes stay in place to reduce movement in the visualization.

In addition to the topology, the GUI includes status indicators for each bed to visualize the internal emergency table. Four labeled rectangles ("BED1" to "BED4") display each bed's real-time heart rate, with background colors indicating the emergency priority.

Heartbeat messages such as "HEARTBEAT BED x BPM: y " are parsed, updating the bed display immediately. During emergencies, the GUI handles EMERG_BEGIN and EMERG_END blocks to update patient priority through color coding. Topology updates are managed using structured TOPO_BEGIN and TOPO_END blocks containing LINK entries.

REFERENCES

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